

Q1.

Hospital patient segmentation.

k	SSE
1	600

2 350

3 220

4 200

5 190.

→ The SSE decreases to 3

After $k=3$, the decrease is verysmall. Therefore the elbow point is $k=3$

optimal number of clusters = 3.

These 3 clusters can help the hospital identify different patient groups.

→ High-risk patients - require closer monitoring and regular treatment.

→ Medium-risk patients - require periodic healthcare support.

→ Low-risk patients - require routine healthcare service.

3 clusters provides meaningful segmentation without creating too many groups.

E-commerce order Analysis.

In hierarchical clustering, orders are grouped based on their similarity in product.

from the dataset :-

- > Electronics :- 01 = 55,000 and 03 = 57,000 $\rightarrow$ very similar.
- $\rightarrow$ clothing :- 02 = 25,000 and 05 = 27,000 $\rightarrow$ very similar
- Furniture :- 04 = 20,000 $\rightarrow$ forms a separate group initially.

These clusters help the retailers understand purchasing patterns. High value electronics customers may be targeted.

Mobile App user behaviour.

In soft clustering, a user can belong to more than one cluster with different membership probabilities.

- 01 : High usage and frequent login $\rightarrow$ mainly highly active.
- 02 : low usage but high purchase $\rightarrow$ may have membership in both low-activity and high value groups.
- 03 : very high usage and login frequency $\rightarrow$ strongly highly active.
- 04 : low usage and low login $\rightarrow$ strongly low activity.
- 05 : High usage and frequent login $\rightarrow$ strongly highly active.

⑤ Employee performance Evaluation.

model SS

$k=2$ 0.48

$k=3$ 0.68

$k=4$ 0.52

The highest Silhouette score is 0.68 for $k=3$.

∴ The most suitable number of clusters is 3.

A possible segmentation is :-

- 1) High-performing Employees - high productivity, attendance and skills.
- 2) Medium-performing Employees - moderate scores.
- 3) Low-performing Employees - lower productivity, attendance and skill scores.

A Silhouette score closer to 1 indicates better defined and well-separated clusters. Since $k=3$ has the highest score.

④ Agricultural crop monitoring.

Before normalization.

- Rainfall: 180-550 mm
- Soil fertility: 3-9
- Crop yield: 65-90 tons.

Because rainfall has much larger numerical values, it can dominate distance calculations in clustering.

Effect of Normalization.

- 1) prevents large-scale features from dominating clustering.
- 2) makes distance calculations fair.
- 3) produces more meaningful clusters.
- 4) Improves comparison between farms.
- 5) makes agricultural analyses more reliable.